Numerical simulation of flow around a two dimensional square cylinder through $\kappa - \omega$ turbulence model

Project results

Fluidodinamica e ingegneria del vento computazionale

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Politecnico di Torino

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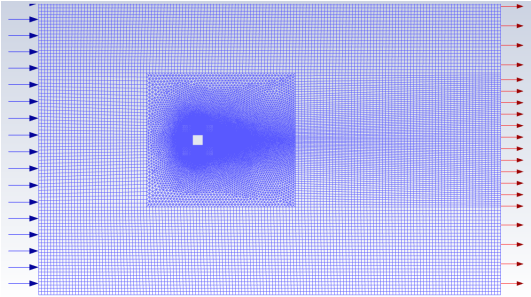
Initial Setup

1 Turbulence model and setting



- $Re = 21400$
- PISO with QUICK
- $It = 2\%$, $L_T = 0.5B \Rightarrow k = 6e - 4$, $\omega = 0.049$
- Domain: $45B \times 15B$ (from $x = -15$ to $x = 30$); Subdomain: $15B \times 13B$
- mesh quad-tri-quad

$\kappa - \omega$ model

1 Turbulence model and setting

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_i}\left[\nu\left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right)\right] - \frac{\partial}{\partial x_j}(\overline{u_i' u_j'})$$

$$\underbrace{\frac{\partial k}{\partial t}}_{T_k} + \underbrace{\frac{\partial(\overline{u_j} k)}{\partial x_j}}_{C_k} = \underbrace{-\overline{u_i' u_j'}\frac{\partial \overline{u_i}}{\partial x_j}}_{P_k} - \underbrace{\frac{\partial}{\partial x_j}\left(\underbrace{\frac{\overline{p' u_j'}}{\rho}}_{a} + \underbrace{\frac{1}{2}\overline{u_i' u_i' u_j'}}_{b} + \underbrace{\nu\frac{\partial k}{\partial x_j}}_{c}\right)}_{D_k} - \underbrace{\nu\overline{\frac{\partial u_i'}{\partial x_j}\frac{\partial u_i'}{\partial x_j}}}_{\varepsilon}$$

$\kappa - \omega$ model

1 Turbulence model and setting

The production of κ through $P_\kappa = -\overline{u_i' u_j'} \dfrac{\partial \bar{u}_i}{\partial x_j}$ is modeled using Boussinesq hypotesis :

$$-\overline{u_i' u_j'} \frac{\partial \bar{u}_i}{\partial x_j} \approx \nu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \frac{\partial \bar{u}_i}{\partial x_j}$$

The diffusion term D_κ is composed by

$$\frac{\partial}{\partial x_j} \left[\frac{\overline{p' u_j'}}{\rho} + \frac{1}{2} \overline{u_j' u_i' u_i'} + \nu \frac{\partial \kappa}{\partial x_j} \right] \approx \frac{\partial}{\partial x_j} \left[\left(\frac{\nu_t}{\sigma_\kappa^\omega} + \nu \right) \frac{\partial \kappa}{\partial x_j} \right].$$

$\kappa - \omega$ model

1 Turbulence model and setting

$$\begin{cases} \dfrac{\partial \kappa}{\partial t} + \dfrac{\partial \bar{u}_i \kappa}{\partial x_j} = \dfrac{\partial}{\partial x_j}\left[\left(\dfrac{\nu_t}{\sigma_\kappa^\omega} + \nu\right)\dfrac{\partial \kappa}{\partial x_j}\right] + P_\kappa - \underbrace{\beta^* \omega \kappa}_{\varepsilon} \\ \dfrac{\partial \omega}{\partial t} + \dfrac{\partial \bar{u}_j \omega}{\partial x_j} = \dfrac{\partial}{\partial x_j}\left[\left(\dfrac{\nu_t}{\sigma_\omega} + \nu\right)\dfrac{\partial \omega}{\partial x_j}\right] + \underbrace{\dfrac{\omega}{\kappa} C_{\omega_1} P_\kappa}_{P_\omega} - \underbrace{C_{\omega_2}\omega^2}_{\varepsilon_\omega} \end{cases}$$

The values of the semi-empirical constants are:

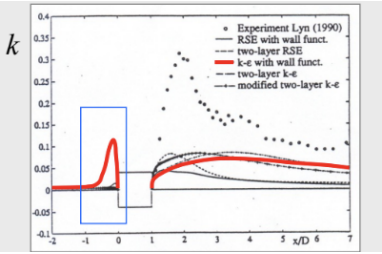
$$\beta^* = 0.09, \qquad C_{\omega_1} = \frac{5}{9}, \qquad C_{\omega_2} = \frac{3}{40}, \qquad \sigma_\kappa^\omega = \sigma_\omega = 2.$$

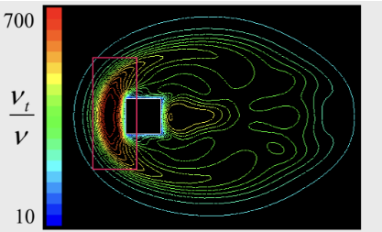
Why $\kappa - \omega$?

1 Turbulence model and setting

$$\frac{\partial \varepsilon}{\partial t} + \frac{\partial \overline{u_j} \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\frac{\nu_t}{\sigma_\varepsilon} + \nu\right)\frac{\partial \varepsilon}{\partial x_j}\right] + \frac{\varepsilon}{k} C_{\varepsilon 1} \cdot P_k - \frac{\varepsilon}{k} C_{\varepsilon 2} \cdot \varepsilon$$

$$\lim_{k \to 0}\left(-\frac{\varepsilon^2}{k} C_{\varepsilon 2}\right) = \infty$$





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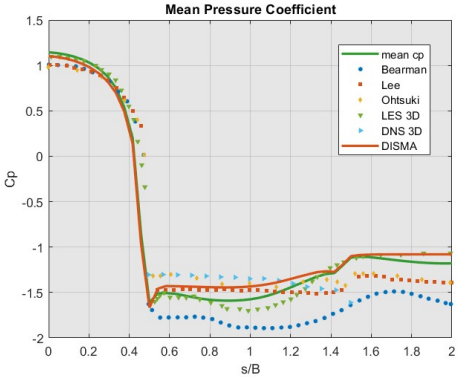
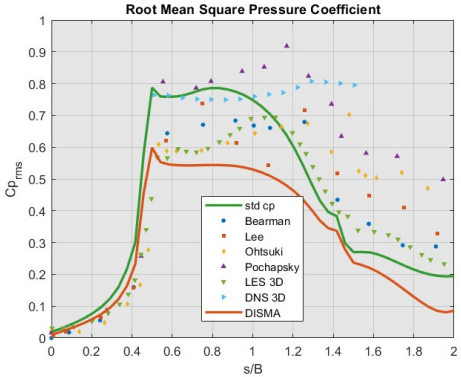
Mean C_p and RMS C_p

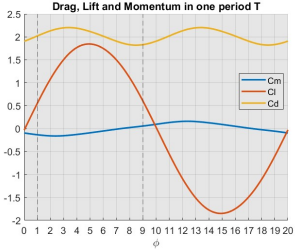
2 Main Result

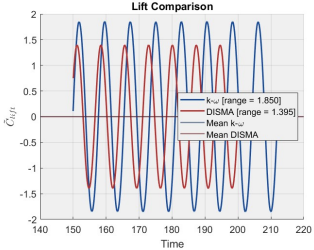


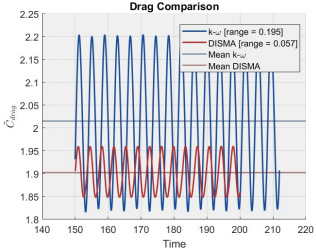


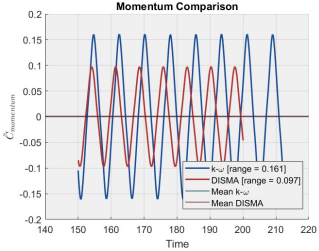
Lift, Drag and Momentum after convergence

2 Main Result









Comparison of Integral values

Authors	St	$\bar{C}_X$	$\tilde{C}_X$	$\tilde{C}_Y$
Lyn	0.135	2.05-2.23	n.a.	n.a.
Sakamoto	0.134	2.22	0.132	0.145
Durao	0.139	n.a.	n.a.	n.a.
Murakami	0.132	2.09	0.13	1.60
DISMA	0.138	1.90	0.057	1.40
$\kappa - \omega$	0.131	1.96	n.a.	n.a.
$\kappa - \omega$ (SST)	0.126	2.13	n.a.	n.a.
$\kappa - \omega$ (our)	0.129	2.02	0.195	1.85

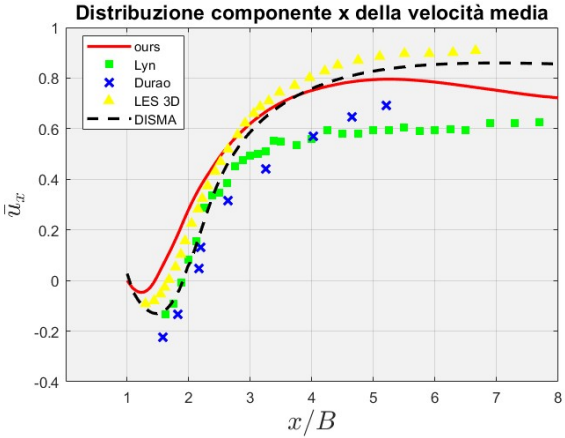
Table: Comparison of Integral values.

Comparison of integral values

Basing on Lyn as reference, with respect to the DISMA:

- Strouhal number is a bit lower
- more accurate on the value of $\bar{C}_x$
- big differences in the oscillation amplitude of $\tilde{C}_x$ and $\tilde{C}_y$
- big differences in C_p rms

Distribution of the x component of mean velocity



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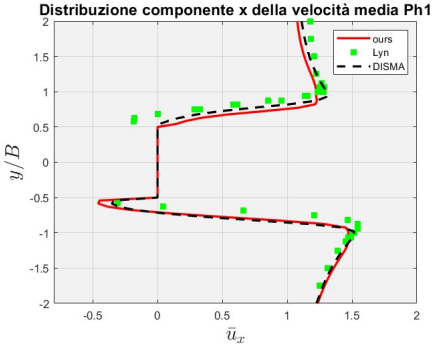
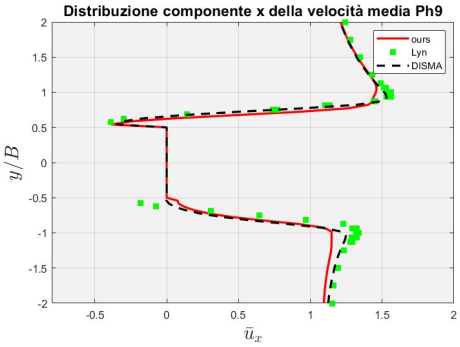
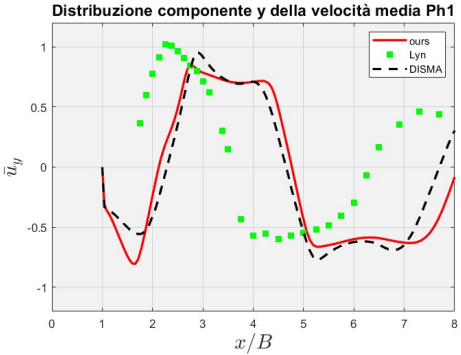
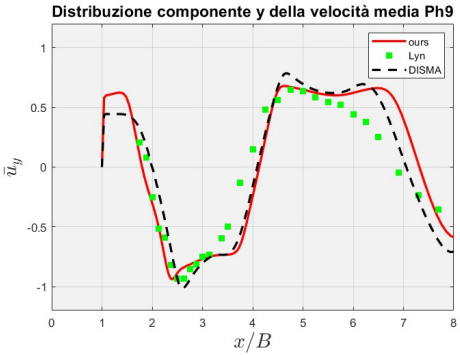




Distribution of the y component of mean velocity





Distribution of the velocities

Basing on Lyn as reference, with respect to the DISMA:

- $\bar{u_x}$ behind the square is not monotone and underestimated in the recirculation region
- $\bar{u_y}$ is overestimated in the recirculation region
- $\bar{u_x}$ along $x/B = 0$ is comparable to DISMA but it is dissimilar to Lyn

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Final Comparison

- $\kappa - \omega$ model predicts mean drag and Strouhal number well.
- Overestimates force fluctuations, underestimates recirculation length.
- $\kappa - \varepsilon$ RNG shows better wake flow prediction under same conditions.
- Improvement possible with SST variant or tuned numerical settings.

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Thank you for listening!

Any questions?

References